

AMENDMENT NO. 1 APRIL 2016
TO
IS 875 (PART 3) : 2015 DESIGN LOADS (OTHER THAN EARTHQUAKE) FOR
BUILDINGS AND STRUCTURES — CODE OF PRACTICE

PART 3 WIND LOADS

(Third Revision)

(Second cover page, Foreword, para 6, line 1) — Substitute ‘discipline’ for ‘descriptive’.

(Second cover page, Foreword, para 6, (c)) — Substitute ‘ k_2 ’ for ‘ k_2 ’.

(Third cover page, Foreword, para after (j), line 8) — Insert ‘anemometer’ after ‘with the help of’.

(Page 2, clause 3.1) — Substitute ‘ C_f ’ for ‘ C_{fd} ’.

(Page 3, Symbol r) — Insert ‘, $I_{h,i}$ ’ after ‘height h ’.

(Page 3) — Substitute ‘ $\bar{V}_{z,d}$ ’ for ‘ $\bar{V}_{d,z}$ ’.

(Page 6, Fig. 1) — Insert the following notes at the end of the figure.

‘NOTES

1 The occurrence of a tornado is possible in virtually any part of India. They are particularly more severe in the northern parts of India. The recorded number of these tornados is too small to assign any frequency. The devastation caused by a tornado is due to exceptionally high winds about its periphery, and the sudden reduction in atmospheric pressure at its centre, resulting in an explosive outward pressure on the elements of the structure. The regional basic wind speeds do not include any specific allowance for tornados. It is not the usual practice to allow for the effect of tornados unless special requirements are called for as in the case of important structures such as, nuclear power reactors and satellite communication towers.

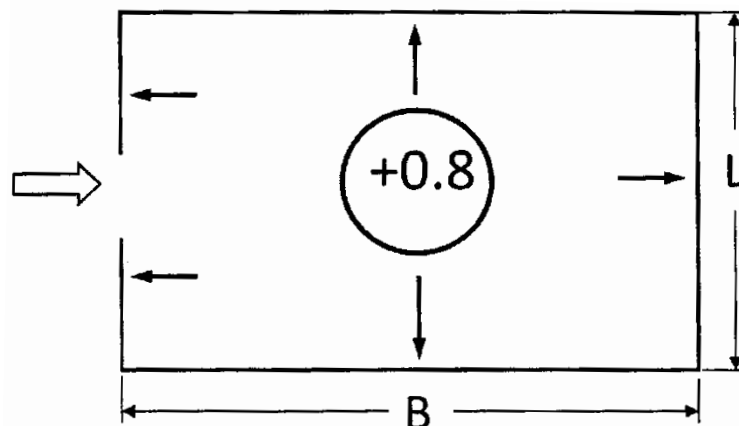
2 The total number of cyclonic storms that have struck different sections of east and west coasts are included in Fig. 1, based on available records for the period from 1877 to 1982. The figures above the line (between the stations) indicate the total number of severe cyclonic storms with or without a core of hurricane winds (speeds above 87 km/h) and the figures in the brackets below the lines indicate the total number of cyclonic storms. These have been included only as additional information.’

*(Page 10, Table 4, *) — Substitute the following for the existing: ‘*Linear interpolation for intermediate values of A is permitted.’.*

(Page 11, clause 7.3.3.3, para 2) — Substitute ‘The solidity ratio Φ ’ for ‘The solidity ratio f ’.

(Page 11, clause 7.3.3.3, para 2, lines 3, 5 and 10) — Substitute ‘ Φ ’ for ‘ f ’.

[Page 12, Fig. 2(b), Right-topmost figure] — Substitute the existing figure with the figure below:



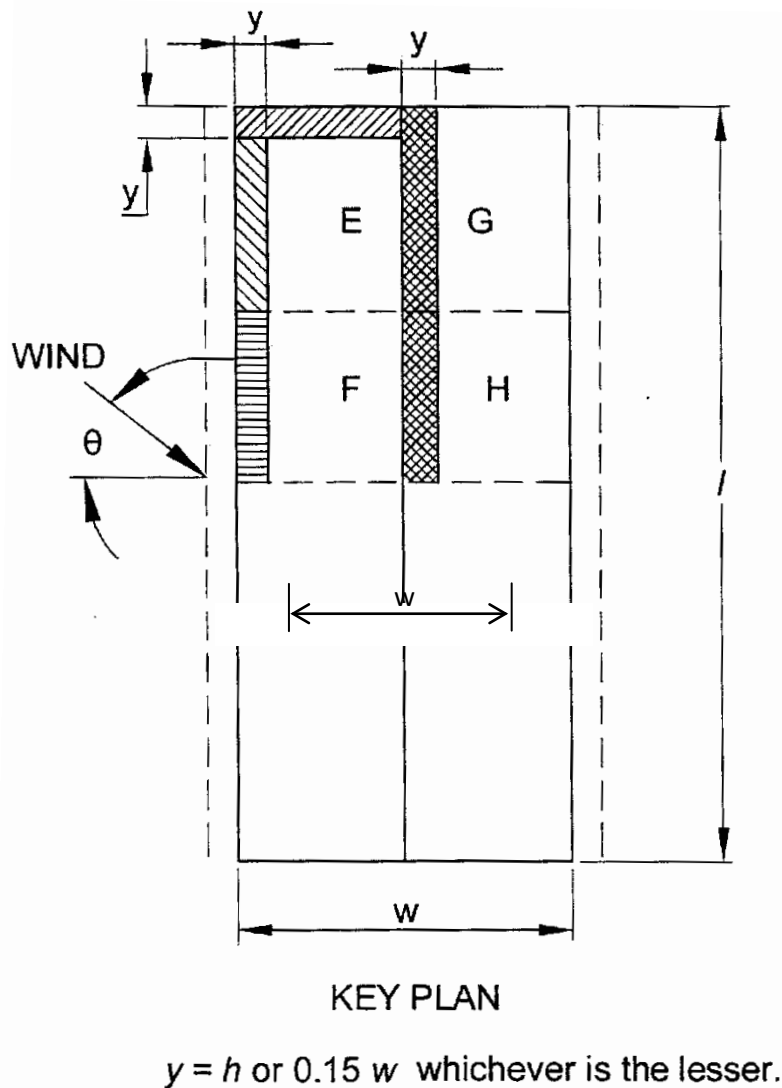
Price Group 4

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(Page 13, Table 5, first row under BUILDING HEIGHT RATIO $\frac{1}{2} < \frac{h}{w} \leq \frac{3}{2}$) — Substitute ' $1 < \frac{l}{w} \leq \frac{3}{2}$ ',
for ' $1 \leq \frac{l}{w} \leq \frac{3}{2}$ '.

(Page 13, Table 5, second row under BUILDING HEIGHT RATIO $\frac{1}{2} < \frac{h}{w} \leq \frac{3}{2}$) — Substitute ' $\frac{3}{2} < \frac{l}{w} < 4$ ', for ' $\frac{3}{2} \leq \frac{l}{w} < 4$ '.

(Page 14, Table 6, bottommost figure) — Substitute the following figure for the existing:



(Page 17, Table 8) — Insert ' α ' after 'ROOF ANGLE'.

(Page 17, Table 8, Note 2) — Substitute 'overhangs.' for 'overhangs,'.

(Page 18, Table 9) — Substitute the following figures for the existing:

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θ	PRESSURE COEFFICIENTS, C_p							
	D	D'	E	E'	END SURFACES			
					C	C'	G	G'
0°	+0.6	-1.0	-0.5	-0.9				
45°	+0.1	-0.3	-0.6	-0.3				
90°	-0.3	-0.4	-0.3	-0.4	-0.3	+0.8	+0.3	-0.4
FOR ALL VALUE OF θ	FOR J : $C_{pTop} = 1.0$, $C_{pbottom} = -0.2$ Tangentially acting friction : $R_{90^\circ} = 0.05 p_d b d$							

(Page 20, Table 11) — Substitute the following for the existing table below figure:

θ	PRESSURE COEFFICIENTS, C_p							
	D	D'	E	E'	END SURFACES			
					C	C'	G	G'
0°	+0.1	+0.8	-0.7	+0.9				
45°	-0.1	+0.5	-0.8	+0.5				
90°	-0.4	-0.5	-0.4	-0.5	-0.3	+0.6	+0.3	-0.4
180°	-0.3	-0.6	+0.4	-0.6				
FOR ALL VALUE OF θ	FOR J : $C_{pTop} = -1.5$, $C_{pbottom} = 0.5$ Tangentially acting friction : $R_{90^\circ} = 0.05 p_d b d$							

(Page 21, Table 12, below figure, line 3) — Substitute ' $\theta = 90^\circ$ ' for ' $\theta = 0^\circ$ '.

(Page 21, Table 12) — Substitute the following for the existing table below figure:

θ	PRESSURE COEFFICIENTS, C_p							
	D	D'	E	E'	END SURFACES			
					C	C'	G	G'
0°	-1.0	+0.3	-0.5	+0.2				
45°	-0.3	+0.1	-0.3	+0.1				
90°	-0.3	0	-0.3	0	-0.4	+0.8	+0.3	-0.6
FOR ALL VALUE OF θ	FOR f : $C_{pTop} = -1.0$, $C_{pbottom} = 0.4$ Tangentially acting friction : $R_{90^\circ} = 0.1 p_d b d$							

(Page 22, Table 13, top figure) — Substitute ' $h'=0.8h$ ' for ' $h'=0.0h$ '.

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(Page 22, Table 13) — Substitute the following for the existing table below figure:

θ	PRESSURE COEFFICIENTS, C_p							
	D	D'	E	E'	END SURFACES			
					C	C'	G	G'
0°	-1.3	+0.8	-0.6	0.7				
45°	-0.5	+0.4	-0.3	+0.3				
90°	-0.3	0	-0.3	0	-0.4	+0.8	+0.3	-0.6
180°	-0.4	-0.3	-0.6	-0.3				
FOR ALL VALUE OF θ	FOR f : $C_{p\text{Top}} = -1.6$, $C_{p\text{bottom}} = -0.9$ Tangentially acting friction : $R_{90^\circ} = 0.1 p_d b d$							

(Page 23, Table 14, top figure) — Delete ‘’.

(Page 24, Table 15) — Delete ‘’.

(Page 24, Table 15, bottom) — Substitute ‘Tangentially’ for ‘Tengentially’.

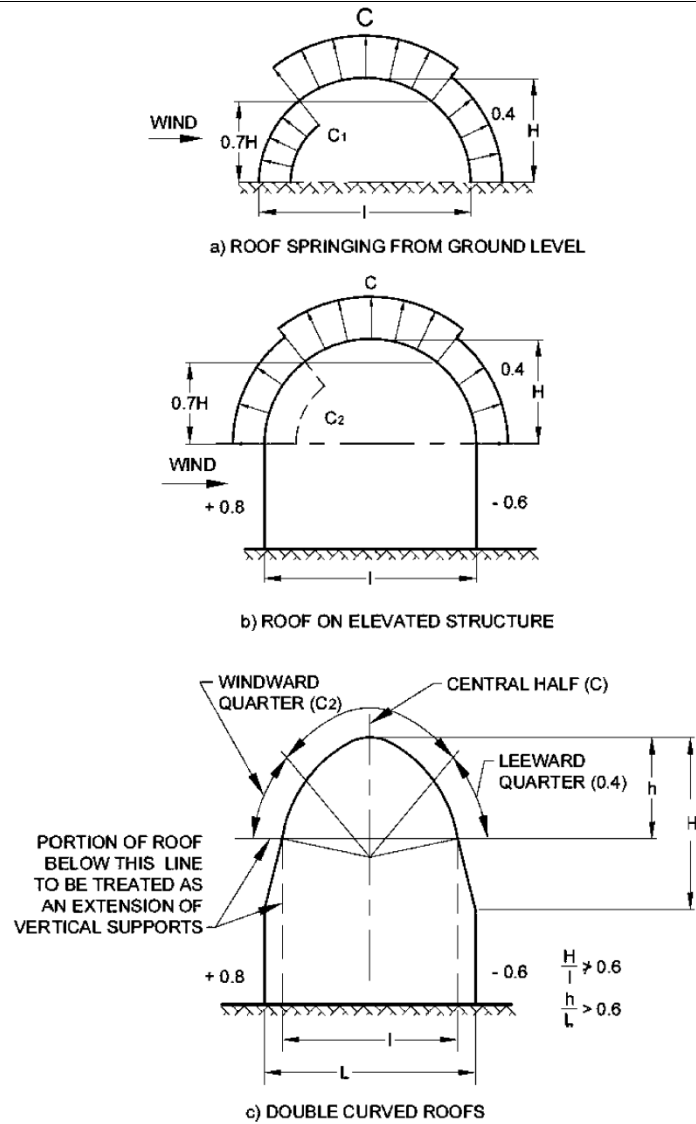
(Page 26, Table 17, bottom) — Substitute the following for the existing entry:

270	Similar to 90° , h_1 , h_2 , h_3 are needed to be reckoned from the windward edge in the same order
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(Page 27, Table 18) — Substitute the following for the existing table:

Table 18 External Pressure Coefficients (C_{pe}) for Curved Roofs
(Clause 7.3.3.6)



VALUES OF C , C_1 and C_2

H/l	C	C_1	C_2	C_2
0.1	-0.8	+0.1	-0.8	+0.05
0.2	-0.9	+0.3	-0.7	+0.1
0.3	-1.0	+0.4	-0.3	+0.15
0.4	-1.1	+0.6	+0.4	-
0.5	-1.2	+0.7	+0.7	-

NOTE: When the wind is blowing normal to gable ends,
 C_{pe} may be taken as equal to -0.7 for the full width of the roof over a length of $l/2$ from the gable ends and -0.5 for the remaining portion.

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(Page 28, Table 19, figures) – Substitute ‘D’ for ‘d’ in both figures.

(Page 31, Table 21, figures, y-axis, between 0.4 and 0.8) – Substitute ‘0.6’ for ‘0.8’.

(Page 31, Table 21) – Substitute the following for the existing entry:

e	See 7.3.3.5
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(Page 32, Table 22) – Substitute the following for the existing entry:

C_{pe}	-0.6	+0.7	See Table 21 for combined roofs
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[Page 34, clause 7.4.1 (a)] — Substitute the following for the existing:

‘a) If $h \leq b$, $F' = C'_f(d - 4h)bp_d + C'_f(d - 4h)2hp_d$, and’

(Page 36, Table 25, Title) — Substitute ‘(Clause 7.4.2.1)’ for ‘(Clause 7.4.2.2)’.

(Page 39, Fig. 5) — Substitute ‘ C_f = FORCE COEFFICIENT’ for ‘ C_f = DRAG COEFFICIENT’.

(Page 39, Table 26) — Substitute ‘FORCE COEFFICIENT, C_f ’ for ‘DRAG COEFFICIENT C_f ’.

[Page 40, Table 28, Sl No. (ii), col 2] — Substitute ‘ $\overline{DV}_d \geq 6 \text{ m}^2/\text{s}$ ’ for the existing information given in bracket.

[Page 40, Table 28, Sl No. (iii), col 2] — Substitute ‘ $b\overline{V}_d \geq 6 \text{ m}^2/\text{s}$ ’ for the existing information given in bracket.

(Page 41, Table 29, Title) — Substitute ‘[Clause 7.4.3.2(a)]’ for ‘[Clause 7.4.3.2 (b)]’.

(Page 42, clause 7.4.3.4, line 10) — Substitute ‘spacing’ for ‘spac-ing’.

(Page 42, Table 30, Title) — Substitute the ‘(l/D = 100)’ in the title in place ‘(L/D = 100)’.

[Page 42, Table 30, Sl No. (i) and (ii), col 2] — Substitute the following for the existing values:

(1)	(2)	(3)	(4)	(5)	(6)
i)	$\overline{DV}_d < 6 \text{ m}^2/\text{s}$	1.2	1.2	1.2	1.3
ii)	$\overline{DV}_d \geq 6 \text{ m}^2/\text{s}$	0.5	0.7	0.9	1.1

(Page 42, Table 31, Force Coefficient C_t for) — Substitute ‘Force Coefficient, C_f for’ for ‘Force Coefficient C_t for’.

(Page 42, Table 31, col 1) — Substitute ‘0.2’ for ‘02’.

(Page 42, Table 31, col 4, sub-sub heading) — Substitute ‘Super Critical Flow ($\overline{DV}_d \geq 6 \text{ m}^2/\text{s}$)’ for the existing.

(Page 43, Table 32, Title) — Substitute ‘ η ’ for ‘ H ’.

(Page 44, Table 34, sub-heading) — Substitute ‘Subcritical Flow ($\overline{DV}_d < 6 \text{ m}^2/\text{s}$)’ for the existing and ‘Supercritical Flow ($\overline{DV}_d \geq 6 \text{ m}^2/\text{s}$)’ for the existing.

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(Page 44, clause **8.2**, line 4) — Substitute ‘x’ for ‘(x)’.

(Page 44, Table 35) — Substitute ‘Subcritical Flow ($\overline{DV}_d < 6 \text{ m}^2/\text{s}$) All Wind Directions’ for the existing and ‘Supercritical Flow ($\overline{DV}_d \geq 6 \text{ m}^2/\text{s}$) All Wind Directions’ for the existing.

(Page 45, Fig.7) — Substitute ‘x’ for ‘X’.

(Page 45, Fig. 8, caption) — Substitute ‘(CLAUSE **8.3**)’ for ‘(CLAUSE 7.3)’.

(Page 46, clause **9.2.1**, Note 2, line 1) — Substitute the following for the existing:

‘2 Unlined welded steel cylindrical structures...’

(Page 46, clause **9.2.1**, Note 4, line 1) — Substitute ‘**9.2.1** (a)’ for ‘**8.2.1** (a)’.

(Page 47, clause 10.2, line 14) — Substitute ‘ $M_a = \sum F_z z$ ’ for ‘ $M_a = \sum F_z Z$ ’.

(Page 47, clause **10.2**, line 28) — Substitute ‘ $(1 + \phi)^2$ ’ for ‘ $(1 + g)^2$ ’ in formula of Gust factor.

(Page 48, clause **10.3**, line 13) — Substitute ‘ $\bar{p}_h$ ’ for ‘ p_h ’.

(Page 49, Fig. 10, x-axis) — Substitute ‘...6 8 10...’ for ‘...6 6 10...’.

[Page 53, Fig.12(b)] — Substitute ‘DESIGN PROFILE AT A’ for ‘OESIGN PROFILE AT A’.

[Page 53, Fig.12(b)] — Substitute ‘WIND DIRECTION’ for ‘WINO OIRECTION’.

[Page 53, Fig.12(c)] — Substitute ‘WIND DIRECTION’ for ‘WINO OIRECTION’.

(Page 54, clause **C-2**, line 6) — Substitute ‘ θ_s ’ for ‘ θ_s ’.

(Page 54, clause **C-2**, line 9) — Substitute ‘distance, X’ for ‘distance, x’.

[Page 54, Fig.13(a)] — Substitute ‘REGION AFFECTED BY TOPOGRAPHICAL FEATURE’ for ‘REGION AFFECTED BY TOPOGRAHICAL FEATURE’.

(Page 55, Fig.14, caption) — Substitute the following for the existing caption:

FIG. 14 FACTOR s_0 FOR CLIFF AND ESCARPMENT’

(Page 55, Fig 15, caption) — Substitute the following for the existing caption:

‘FIG. 15 FACTOR s_0 FOR RIDGE AND HILL’

(Page 56, Fig.17) — Substitute ‘ s_p ’ for ‘sp’.

(Page 56, clause **D-1**, para 7, line 5) — Substitute ‘ s_p ’ for ‘sp’.